

Capstone Weekly Status Report 13

Date and Time: November 30, 2025

Student Name: Satwik Alla

Capstone Project: SBOM Manager - Software Bill of Materials Management System

Capstone Sponsor: Software Engineering Department **Ethical Software Lab** – In-house project

Capstone Advisor: Prof. Mohammad Samarah & Prof. Melissa Sahl

Status Details:

Work accomplished this Week

Statistics Dashboard Implementation

- Completed comprehensive statistics page with platform distribution, component analytics, license charts, and storage usage tracking
- Integrated Chart.js for interactive data visualizations
- Added SBOM generation success/failure rate metrics

Vulnerability Integration

- Researched and tested NVD API integration with CVE data fetching
- Implemented vulnerability matching algorithm and database schema extensions
- Added caching strategy to handle API rate limits (5 requests/30 seconds)

Enhanced User Experience

- Implemented notifications for SBOM completion
- Improved search autocomplete with debouncing
- Enhanced mobile responsiveness across all pages

Work Planned for Next Week

Post-Beta Improvements & New Features

- Addressed Beta checkpoint feedback and implemented suggested improvements
- Added progress indicators for large file processing and advanced filtering options (platform, license type, component count)
- Implemented bulk operations for managing multiple applications simultaneously
- Enhanced error messaging and recovery mechanisms

Final Deliverables

- Complete Final Capstone Report (40-50 pages) with executive summary
- Create final presentation slides and demo video (10-15 minutes)
- Prepare all 13+ final checkpoint deliverables for submission

Final Features & Polish

- Complete vulnerability integration with automated CVE alerts
- Finalize Docker containerization with pre-installed Syft
- Implement real-time WebSocket updates (if time permits)
- Complete end-to-end testing suite

Final Validation

- Comprehensive security and performance testing
- Cross-browser compatibility verification
- User acceptance testing with external testers
- Final code cleanup and documentation

Problems, obstacles, needs, or questions that I need help with from client, advisor, or instructor

None

Overall Status

- Project Status Color: Green
- Project Status Color from Previous Week: Green